Cohort build log (v2 rebuild)

Running record of how each cohort was built, the decisions taken and the numbers produced. Source for the methods section, the flow diagram and the cohort table.

TCGA-LUAD (scripts/10_cohort_tcga.R) — built 19 Oct 2026

- Source: GDC STAR-Counts (GENCODE v36) via TCGAbiolinks 2.40.0; tpm_unstranded, log2(TPM + 1); primary tumours, first barcode per patient. Library type: poly(A) selection.
- Result: 504 patients, 182 deaths (matches the corrected count in the v0.3 HANDOVER).
- A transient "Failure when receiving data from the peer" warning appeared during the GDC metadata query; the cohort was built normally.

CPTAC-3 lung adenocarcinoma (scripts/11_cohort_cptac3.R, R/gdc_api.R) — built 19 Oct 2026

- Why not TCGAbiolinks: GDCquery() stops on CPTAC-3 with "object 'sample_type' not found" (the GDC split sample_type into tissue_type, tumor_descriptor, specimen_type and preservation_method; some CPTAC-3 samples carry only the new fields). Its barcode filter also compares against aliquot IDs, so it could not have matched CPTAC-3 case IDs. CPTAC-3 is therefore read directly from the GDC API.
- GDC Data Release 46.0 (10 Aug 2026).
- Cases: 1,866 CPTAC-3; 339 lung; 230 lung adenocarcinoma (primary_site "Bronchus and lung" and disease_type "Adenomas and Adenocarcinomas"); all "Adenocarcinoma, NOS", one diagnosis each.
- RNA-seq files of these cases: 446 (215 normal, 3 metastatic, 228 primary tumour). 50 primary-tumour files list several samples of the same patient (one RNA extract from several tumour pieces; the CPTAC LUAD study cryopulverised specimens and aliquoted the material); none was pooled across patients. Rule: a file is eligible if it comes from one case and every sample is a primary tumour; one file per patient (first by sample ID, then file ID).
- Patients with RNA-seq: 228; with valid survival: 205; deaths: 55. Exclusions for survival are mainly vital status "Not Reported" (22 of 230 cases).
- Last follow-up: latest follow_ups $days_to_follow_up (the rule TCGAbiolinks applies to TCGA); $diagnoses days_to_last_follow_up only as a fallback. Checked: the two sources are identical for 204 of 205 analysed patients; the one difference (63 days) is a patient who died and has a follow-up record dated after death, which does not affect overall survival (death date used).
- Age: age_at_index is filled for only 2 of 230 cases; age is taken from diagnoses$age_at_diagnosis (days, converted to completed years) for 225, so age is recorded for all 205 analysed patients. This keeps age in the H2 model (used when at

least 80% recorded), as for TCGA.

- Stage: I 114, II 46, III-IV 45 (all staged). Sex: 131 male, 74 female. Median follow-up (reverse Kaplan-Meier) 35.6 months; maximum 72.
- Change from v0.3: the old CPTAC-3 cohort had 150 patients and 38 deaths; the old filter most likely dropped the files that list several tumour samples. All CPTAC-3 numbers in the manuscript will change.

RNA-seq library type and TPM composition (TCGA-LUAD vs CPTAC-3) — 19 Oct 2026

- CPTAC LUAD libraries: TruSeq Stranded Total RNA with Ribo-Zero Gold (total RNA, rRNA-depleted; Gillette et al., Cell 2020, doi:10.1016/j.cell.2020.06.013). TCGA: poly(A) selection. Many lncRNAs are not polyadenylated or stay in the nucleus, so the two library types capture them differently: a cohort-level source of heterogeneity for the Discussion and the cohort table.
- Share of TPM outside protein-coding and lncRNA genes, per sample (min / Q1 / median / Q3 / max): TCGA-LUAD 0.017 / 0.062 / 0.095 / 0.173 / 0.748; CPTAC-3 0.273 / 0.367 / 0.413 / 0.468 / 0.672.
- Median log2(TPM + 1), TCGA vs CPTAC-3: RN7SL1 9.19 vs 17.81; MT-RNR2 13.17 vs 12.63; GAPDH 10.49 vs 9.73; ACTB 12.01 vs 10.74.
- Within-cohort z-scoring removes the constant shift but not the sample-to-sample part (interquartile spread about 0.18 log2 units in TCGA and 0.25 in CPTAC-3; the most extreme TCGA samples are about 2 log2 units low).
- Decision: primary analysis unchanged (GDC TPM as provided). Pre-specified sensitivity analysis added to the kit (config sensitivity$tpm_rescale_gene_types; rescale_tpm_cohort(); setting in scripts/34_sensitivity.R; QC columns in scripts/19_cohort_qc.R): RNA-seq TPM rescaled so that protein-coding and lncRNA genes (GENCODE v36) sum to 10^6 per sample before log2(TPM + 1) and z-scoring. Must be written into the OSF amendment before any outcome analysis. No sample exclusion based on this share (it would be data-driven); the share is reported in QC.

GEO microarray cohorts (scripts/12_cohort_gpl570.R, config/cohorts.yaml) — settings fixed 19 Oct 2026

Fields confirmed against logs/geo_fields.txt (show_geo_fields output). An event value that is neither dead nor alive (e.g. "NTL", "n/a") is coded missing, never alive.

- GSE31210: tissue "primary lung tumor" (226 of 246; 20 normal lung excluded). OS = "days before death/censor" (days); event "death" dead/alive (35 dead, 191 alive). Stage "pathological stage" IA/IB/II. Age from "age (years)", completed from "age" (GEO splits age over two fields; the second holds 5 samples). Sensitivity flag: "exclude for prognosis analysis due to incomplete resection or adjuvant therapy" = "exclude" (22 cases).
- GSE50081: histology "adenocarcinoma" only (127 of 181; adenosquamous, large cell, "NSCLC-favor adenocarcinoma" excluded). OS = "survival time" (years); event "status" dead/alive. Stage "Stage" is written 1A/1B/2A/2B and is recoded to IA/IB/IIA/IIB (without

the recode every stage would have been missing). Age "age"; sex "Sex".

- GSE30219: histology "ADC" (85 of 307). OS = "follow-up time (months)"; event "status" DEAD/ALIVE. No stage field: stage group derived from pT/pN/pM with AJCC 6th-edition grouping (T not split into a/b in this series): I = T1-2 N0 M0; II = T1-2 N1 M0 or T3 N0 M0; III-IV = T3 N1-2, T4, N2-3 or M1; MX or missing M counted as M0 (resected series); TX/NX give missing stage unless III-IV anyway. Limitation to state: under AJCC 7th edition some T2 N0 tumours (>5 cm) would be stage II.
- CEL archives: downloaded under a temporary name and kept only when the size matches the server's; RMA stops if any selected sample lacks a CEL file. BioMart requests (Ensembl release 114 probe map) are retried.
- First run of script 12 (19 Oct 2026): GSE50081 127 patients, 51 deaths, median follow-up 65.9 months, stage I 92 / II 35; GSE30219 83 of 85 with valid survival, 43 deaths, median follow-up 116 months, but stage missing for all 83. Cause: the R yaml package reads an unquoted key `n` as FALSE (YAML 1.1 booleans y/n/yes/no/on/off), so `stage_tnm: {t, n, m}` lost the N field. Fixed by naming the keys tumour / node / metastasis; make_clin_geo() now stops if any is missing, and the config test reads the real YAML file. Rule for all config files: no bare y/n/yes/no/on/off keys or values.
- GPL570 probe map: 54,675 probes; 20,274 Ensembl genes kept (one probe per gene).
- Accepted (19 Oct 2026): GSE31210 226 patients, 35 deaths, median follow-up 60.7 months, stage I 168 / II 58, age for all 226 (two GEO age fields combined), 22 flagged for the sensitivity analysis; GSE50081 127 patients, 51 deaths, median follow-up 65.9 months, stage I 92 / II 35 (1A 36, 1B 56, 2A 7, 2B 28). Expression range 1.85-15.08 (RMA log2).
- GSE30219: 85 adenocarcinomas, 83 with valid survival, 43 deaths, median follow-up 116 months. Follow-up is recorded in whole months; the 2 excluded patients have follow-up 0 and both died (deaths in the first month after surgery are coded 0 and fall under the pre-specified rule follow-up > 0): state in the flow diagram. Stage after the fix (20 Oct 2026; 85 selected): T1N0M0 71 and T2N0M0 10 -> I; T2N1M0 2 and T3N0M0 1 -> II; T3N1M0 1 -> III-IV. Analysed 83: I 79, II 3, III-IV 1. Stage is nearly constant in GSE30219 (and GSE31210 / GSE50081 hold only stage I-II by design), so stage adjustment in H2 carries little information in the microarray cohorts: state this when reporting H2.
- GSE81089 (RNA-seq, 199 NSCLC + 19 paired normal; Uppsala University Hospital, surgery 2006-2010; clinical data from the regional lung cancer registry). Codebook from the series matrix header: histology 1 = squamous cell, 2 = adenocarcinoma ("AC unspecified"), 3 = large cell/NOS (2 = 108 samples); stage (pTNM, reviewed 2013) 1 = IA, 2 = IB, 3 = IIA, 4 = IIB, 5 = IIIA, 6 = IIIB, 7 = IV; dead 0 = no, 1 = yes; vital date = date of death or latest contact; surgery date = date the sample reached pathology; age at surgery; smoking 1 = current, 2 = ex (>1 year), 3 = never. Overall survival = vital date minus surgery date. Expression: GSE81089_FPKM_cufflinks.tsv.gz and GSE81089_readcounts_featurecounts.tsv.gz (columns L608T_2122 and L771T_1 belong to samples L608T and L771T). The v0.3 pipeline dropped this cohort because of an identifier bug (R9), not missing data; restoring it requires the OSF amendment. Library (Djureinovic et al., JCI Insight 2016, doi:10.1172/jci.insight.86837): Illumina TruSeq RNA

v2 with poly(A) selection (as TCGA), HiSeq 2500, 2 x 100 bp; TopHat 2.0.8b on GRCh37 with Ensembl 73 genes; FPKM from Cufflinks 2.1.1. FPKM file: 63,130 Ensembl genes (no version suffix) x 218 samples named as in the GEO field "tumor (t) or normal (n)". Script 13: adenocarcinomas (histology 2), TPM = FPKM / sample total x 1e6, log2(TPM + 1); follow-up from the two dates. Ensembl 73 is older than GENCODE v36, so newer lncRNA IDs will be absent: coverage will show it. Built for description; enters the analyses only after the amendment. First run stopped: the FPKM file contains missing values. Rule added: genes with a missing FPKM in any selected adenocarcinoma are dropped (they could not pass the presence rule anyway) and TPM is computed over the remaining genes; the script stops if any sample has more than 5% missing. Second run stopped on negative values: all 579 negative FPKM values among the adenocarcinomas are exactly -1 (a code for "not quantified"; at most 13 per sample, at most 52 samples per gene), so -1 is treated as missing; any other negative value stops the script. Dropped genes are listed in data/processed/GSE81089_genes_dropped_missing.csv, to check against the signature genes at the coverage step. Counts reported in results/qc/GSE81089_build_summary.txt.

- GSE81089 built (20 Oct 2026): histology codes of all samples 1 = 67, 2 = 108, 3 = 24, missing 19 (normals). 108 adenocarcinomas: 49 dead, 59 alive; stage codes 1:44, 2:18, 3:8, 4:11, 5:24, 7:3. FPKM file 63,130 genes x 218 samples; among the adenocarcinomas 579 values coded -1 (75 genes) and 1 blank (in a non-Ensembl row "TC%..."); 76 rows dropped, 63,054 Ensembl genes kept (all kept rows are Ensembl genes, so the stray row never entered the TPM totals). Analysed: 107 patients (1 excluded: follow-up 0, died on the day of surgery), 48 deaths, median follow-up 52.3 months (registry extraction dates 28 Apr 2013 and 10 Apr 2013 dominate the last-contact dates), stage I 61 / II 19 / III-IV 27, age for all, 69 female / 39 male. Expression range 0-17.62. Listed in config as cohorts$candidate (QC and coverage only) until the amendment adds it.

All cohorts built (20 Oct 2026)

Cohort	Role	Platform / library	Patients	Deaths	Median follow-up (months)
TCGA-LUAD	reference (apparent performance)	RNA-seq, poly(A)	504	182	29.0
CPTAC-3	independent	RNA-seq, total RNA (Ribo-Zero)	205	55	35.6
GSE31210	independent	GPL570, RMA	226	35	60.7
GSE50081	independent	GPL570, RMA	127	51	65.9
GSE30219	independent	GPL570, RMA	83	43	116
GSE81089	independent (restored; must be named in the amendment)	RNA-seq, poly(A), Ensembl 73	107	48	52.3

Independent deaths:
184 without
GSE81089, 232 with it.

- Open gap: config lists GSE72094 as coverage_only, but no script builds it. Its coverage can be judged from the GPL15048 platform annotation; to be added at the coverage step (Step 3).

Cohort QC (scripts/19_cohort_qc.R) — 20 Oct 2026

- All six cohorts pass check_cohort(). Sample flow (with expression / with clinical record / with valid survival): TCGA-LUAD 517 / 517 / 504; CPTAC-3 228 / 228 / 205; GSE31210 226 / 226 / 226; GSE50081 127 / 127 / 127; GSE30219 85 / 85 / 83; GSE81089 108 / 108 / 107.
- Stage I-II: TCGA 78.6% (8 missing), CPTAC-3 78.0%, GSE31210 100%, GSE50081 100%, GSE30219 98.8%, GSE81089 74.8%. Age recorded: TCGA 98%, all others 100%.
- Share of TPM outside protein-coding and lncRNA genes (median / max): TCGA 0.095 / 0.748; CPTAC-3 0.413 / 0.672; GSE81089 0.251 / 0.555 (Ensembl 73 IDs matched to GENCODE v36 gene types, so retired IDs count as "other").
- Expression boxplots: the three GPL570 cohorts are aligned after RMA; RNA-seq panels show all ~60,000 genes (medians near 0) and nothing abnormal; one GSE81089 sample has a lower upper quartile, consistent with the composition differences.
- Decision (20 Oct 2026): GSE81089 included as an independent cohort (config cohorts$independent; script 13 in run_all.R). The OSF amendment must name it before scripts 30-34 run, and independence.csv must be completed for it.

- Project folder moved by the user to ~/Desktop/Lab Works/MHENCR/luad-lncrna-benchmark-v2 (20 Oct 2026). All paths in the pipeline are relative to the project root, so nothing in the code changes.

After OSF Update 3 (filed 20 Oct 2026)

- OSF Update 3 filed and committed by the user on 20 Oct 2026 (field-by-field text: claude/osf-update3-final-paste-text.md).
- Project folder moved by the user to ~/Desktop/Lab Works/BenchMark3/luad-lncrna-benchmark-v2 (20 Oct 2026). Paths in the pipeline are relative, so no code changes.
- Code update v2-update-D2-D4 (20 Oct 2026):
  - D2: stage coded in four levels I / II / III / IV everywhere (collapse_stage(); stage_from_tnm() puts M1 in IV, other III-defining TNM in III).
  - D4: TCGA-LUAD keeps, per patient, the primary tumour with the largest total read count (sum of STAR unstranded counts over Ensembl genes); ties by barcode. choose_tcga_samples(); choices written to data/processed/TCGA-LUAD_sample_choice.csv. Script 10 reuses the saved SummarizedExperiment and clinical data (no new download) and keeps the GDC release of the first download.
  - CPTAC-3 frozen: script 11 reuses the saved GDC answers (cases_api.rds, files_api.rds) so the cohort stays at Data Release 46.0; querying again would be a reportable deviation.
  - Script 21 downloads GENCODE v22 from the EBI GENCODE archive if missing; *.gtf.gz kept out of git.
  - independence template extended with GSE72094 rows (15 x 7 cohorts).
  - All cohorts must be rebuilt once (scripts 10, 11, 12, 13, 19) because stage is stored in the cohort objects. Patients and deaths are expected unchanged; only stage III/IV split and the TCGA sample choice change.